Aircraft Engine Remaining Useful Life (RUL) Prediction Using Machine Learning

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***Abstract*—Knowing the Remaining Useful Life (RUL) of aircraft engines is of paramount importance in the aviation industry. RUL helps anticipate engine failures beforehand so that airlines can proactively schedule maintenance, optimize resource allocation, and reduce the risk of downtime. In this work, we consider two problems on the prediction of RUL: a binary classification problem to predict whether the engine will fail within a month, and a regression problem to predict the remaining number of operational cycles before engine failure. To this end, using the NASA C-MAPSS dataset, we trained a slew of machine learning models to address the aforementioned two problems. Our results show that the Long Short-Term Memory (LSTM) model performed the best on the binary classification problem with 0.95 precision, 0.88 recall and 0.91 F1-score on the test set, and that the Convolutional Neural Network (CNN) model [1] best on the regression problem with 14.02 RMSE on the test set. The state-of-the-art paper documented an RMSE of 16.42. Our work not only surpassed the reference RMSE but also demonstrated superior predictive accuracy. Comparing these results highlights the substantial progress achieved in predicting aircraft engine Remaining Useful Life, showcasing the effectiveness of the models developed in this study in outperforming the state-of-the-art benchmark.**

I. Introduction

The aviation industry faces a critical challenge: ensuring the safety and efficiency of operations while minimizing the risk of costly and potentially dangerous engine failures. Traditional maintenance approaches, often based on fixed schedules or simple usage thresholds, prove inadequate in addressing this challenge. They can lead to unscheduled downtime, unnecessary maintenance actions, and inefficient resource allocation, ultimately impacting safety, operational performance, and profitability.

The emergence of accurate Remaining Useful Life (RUL) prediction offers a promising solution. By anticipating engine failures before they occur, airlines can proactively schedule maintenance, optimize resource allocation, and reduce the risk of downtime. This translates to significant benefits, including:

- Enhanced Safety: Proactive maintenance based on RUL predictions helps prevent catastrophic failures, safeguarding passengers and crew.
- Optimized Maintenance Scheduling: Knowledge of RUL enables precise scheduling of maintenance tasks, maximizing aircraft availability and minimizing costly downtime.
- Reduced Maintenance Costs: By prioritizing critical repairs and avoiding unnecessary interventions, RUL predictions lead to significant cost reductions in maintenance operations.
- Extended Engine Life: Operating engines within optimal conditions based on RUL estimation allows airlines to maximize the lifespan of their engines, ensuring a greater return on investment.

Several machine-learning techniques have been explored for RUL prediction, including regression models, damage propagation models, and deep convolutional models. Each category offers unique advantages, with deep learning models showcasing the potential for superior accuracy. However, the effectiveness of any approach relies on the quality of the data and the selection of relevant features.

This paper delves into the development and evaluation of robust RUL prediction models for aircraft engines. It employs the NASA C-MAPSS dataset [2], undergoes data preprocessing step, and applies a piece-wise RUL function to generate the RUL column to be predicted. By implementing and comparing various machine learning models, the project identifies the most accurate and generalizable approach for predicting RUL.

Preliminary results demonstrate the effectiveness of the proposed methodology. Deep learning models, particularly Recurrent Neural Networks and Convolutional Neural Networks exhibit superior accuracy in predicting RUL compared to other models.

In the following sections, we discuss related works, the methodology employed in this study, detailed results, and analysis, as well as the conclusive findings.

II. Related Works

The exploration of RUL estimation for aircraft components has seen significant contributions from various studies, each addressing specific challenges and introducing novel methodologies.

Saxena [3] presented a pioneering approach in "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation." This study focused on modeling damage propagation within aircraft gas turbine engine modules. By employing a thermo-dynamical simulation model, the authors generated response surfaces for all sensors, simulating faults with worsening effects and providing valuable challenge data for the PHM'08 competition. This study laid a foundation for understanding and generating the train, test, and validation data sets for the development of data-driven prognostics.

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In response to the limitations of linear models, the study by Zheng [4] proposed an approach based on the Recurrent Neural Networks for Remaining Useful Life Estimation. Recognizing the critical role of accurate RUL estimation in Prognostics and Health Management (PHM), the authors introduced a Long Short-Term Memory (LSTM) model [5]. This method leveraged sequence information for improved accuracy, showcasing superior performance over traditional regression-based and Convolutional Neural Network (CNN) approaches. The study's application across three widely adopted PHM datasets demonstrated the LSTM model's efficacy, emphasizing its potential in the expanding realm of Industrial Internet of Things (IoT) applications.

Building on these works, the paper by Lim and Fellows [6] introduced an extension to an existing Kalman Filter (KF) ensemble method to estimate the RUL. Recognizing the limitation of the KF ensemble in handling non-linear models, the authors proposed the Switching Kalman Filter (SKF) to approximate non-linear functions with piece-wise linear functions. This approach allowed the SKF ensemble to choose the most probable degradation mode during RUL estimation, showcasing effectiveness in detecting switching points and outperforming existing methods on NASA's C-MAPSS Dataset and the PHM 2008 Data Challenge Dataset [7]. This study demonstrated the potential of the SKF ensemble in addressing non-linear degradation patterns and broadening its application to various prognostic situations.

In a parallel development, the study on the Gated Recurrent Unit Networks for Remaining Useful Life Prediction [8] provided a comprehensive approach to RUL prediction. This paper introduced a novel methodology based on Gated Recurrent Unit (GRU) networks [9], emphasizing automatic feature extraction from time series data. The proposed approach demonstrated superior prediction performance on NASA's commercial modular aero-propulsion system simulation data, showcasing the effectiveness of GRU networks in achieving accurate RUL estimates.

Together, these studies contribute to the advancement of RUL estimation techniques, addressing challenges posed by linear models and emphasizing the importance of sequence information and automatic feature extraction in enhancing predictive accuracy for aircraft components.

III. Methodology

A. Problem Framing

In this study, the exploration of predictive models for aircraft engine RUL involved two distinct problem formulations, each tailored to address specific operational requirements and decision-making contexts.

1) Binary Classification Problem: The first problem frame revolves around a binary classification task, aiming to predict whether an engine is likely to fail within the next 30 days or not. This formulation provides a binary outcome, simplifying the prediction into a critical operational decision – identifying whether an engine requires immediate attention due to an imminent risk of failure. The model is trained to distinguish between two classes: "Failure within 30 Days" and "No Failure within 30 Days."
2) Regression Problem: The second problem frame approaches the RUL prediction as a regression problem, focusing on estimating the remaining number of operational cycles before engine failure. This formulation provides a continuous numerical prediction, offering a detailed understanding of the remaining engine lifespan. The model is trained to learn the complex relationships between input features and the actual Remaining Useful Life, providing a quantitative estimate that facilitates proactive maintenance planning.

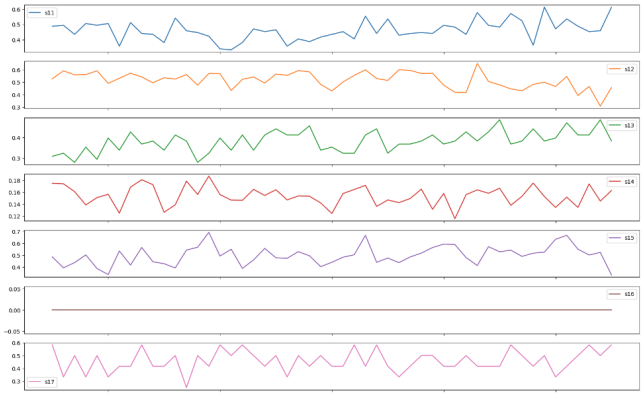

Fig. 1. Sensor Readings.

Each problem formulation serves a specific purpose, catering to different decision-making scenarios. The binary classification problem is geared towards timely and actionable insights, while the regression problem provides a detailed quantitative forecast.

The subsequent sections detail the data preprocessing, feature engineering, and model selection steps undertaken for each of these problem formulations, offering a comprehensive overview of the methodology employed in this study.

B. Dataset

The dataset utilized in this study is sourced from NASA and encompasses multivariate time series data from 100 unique aircraft engine units. Each unit's operational behavior is captured through readings from 21 sensors and observations of 3 distinct operational settings. These engines, belonging to the same type, initiate their respective time series with varied levels of initial wear and manufacturing differences. Figure 1 visualizes the reading from some of the selected sensors included in the dataset.

Within each dataset instance, critical information regarding three operational settings known to influence engine performance is embedded. To emulate real-world conditions, the dataset introduces intentional sensor noise, effectively sim-

ulating uncertainties and imperfections inherent in practical aviation situations.

Regarding the time series structure, each sequence commences with the engine operating under normal conditions, devoid of any faults or performance issues. Subsequently, the engines undergo a degradation process at some point during the time series. Notably, in the training set, this degradation intensifies until it reaches a predefined threshold, indicating a point at which continued operation is not recommended. Importantly, the test set encapsulates a practical scenario by concluding the time series before the engine reaches complete degradation.

C. Modeling and Training

In the pursuit of accurate RUL estimation for aircraft engines, a diverse set of models was constructed and trained on the provided dataset. These models encompassed three distinct categories, each leveraging different techniques to capture and predict the underlying patterns in the data.

1) Regression Models: The first category of models explored traditional regression techniques to understand the linear relationships within the dataset. The following regression models were trained and tested:

- Linear Regression [10]
- Random Forest [11]
- K-Nearest Neighbors [12]
- XGBoost [13]
- Support Vector Machine [14]
- Decision Tree [15]
- Extra Trees Regression [16]
- AdaBoost Regression [17]

These models were chosen to assess the dataset's responsiveness to various regression methodologies, each with its unique strengths and assumptions.

2) Deep Recurrent Models: Recognizing the temporal nature of the data, the second category comprised sequence-based models that could effectively capture time-dependent patterns. The following models were trained in this category:

- Recurrent Neural Network (RNN)
- Long Short-Term Memory (LSTM)
- Bidirectional LSTM
- Gated Recurrent Unit (GRU)

This category aimed to exploit the sequential dependencies within the dataset, allowing the models to learn and adapt to the changing patterns over time.

3) Deep Convolutional Models: The third and final category delved into the realm of Convolutional Neural Networks, with both 1D and 2D architectures being employed. These models were designed to compare their performance in recognizing patterns as compared to the traditional regression or sequence-based models. The three types of CNNs we considered are listed in the following, where numbers represent the amount of convolutional and pooling layers.

- CNN-2
- CNN-4
- CNN-1+LSTM

Throughout the modeling process, cross-validation was employed to optimize the models' performance. The training data were used to fine-tune each model, striking a balance between model complexity and predictive accuracy. The evaluation metrics included standard regression metrics such as Mean Squared Error (MSE) and Root Mean Squared Error (RMSE), providing a comprehensive understanding of each model's predictive capabilities.

By exploring these diverse model architectures, this study aimed to understand the most effective approaches for RUL prediction, considering both the temporal aspects and intricate dependencies present in the dataset. The subsequent sections will delve into the results obtained from these trained models and their implications for effective prognostics and health management of the aircraft engine.

IV. Results and Analysis

A. Experiment 1: Dataset Analysis

The goal of Experiment 1 was to investigate the impact of using three different types of datasets on the performance of machine learning models. Three datasets were created, each subject to multiple machine learning (ML) models. The datasets were as follows:

1) **Dataset 1:** All 25 columns, including 21 sensors, 3 operational settings, and the current cycle.
2) **Dataset 2:** Excluded 6 columns with constant values.
3) **Dataset 3:** Excluded the constant sensors and the 3 operational settings.

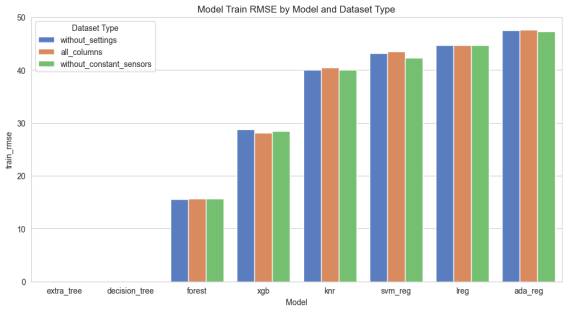

Fig. 2. RMSE of Different Models for Different Datasets

The RMSE for each ML model on each dataset was plotted, as shown in Figure 2. Surprisingly, the performance of different models on these datasets remained constant, and no significant improvements were observed for any dataset. This indicated that the choice of columns did not have a substantial impact on the model performance.

Consequently, this observation established a baseline for the subsequent experiments. The dataset with all columns (Dataset 1) was selected for further analysis, as the variations in column selection did not yield discernible improvements in model performance.

B. Experiment 2: Classification

In Experiment 2, various machine learning models were constructed to address the binary classification problem using the dataset. The primary objective was to predict whether the engine would fail within one month or not (binary classification). Figure 3 shows the learning curves, including the loss and accuracy of the LSTM binary classifier during training.

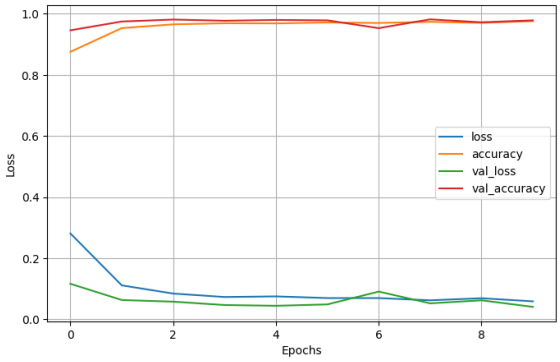

Fig. 3. Training vs Validation Loss and Accuracy for LSTM Classifier

In the classification experiment, the focus was solely on evaluating the efficiency of the Long Short-Term Memory (LSTM) model in predicting engine failure within one month. The optimal LSTM configuration involved two layers: the initial layer comprised 100 units, followed by a second layer with 50 units. A dropout layer with a rate of 0.2 was applied between each of these layers. The classifier achieved impressive performance metrics: 96% on precision, 0.88 recall, and 0.92 F1-score.

C. Experiment 3: Regression

Experiment 3 was dedicated to training models for the regression task, specifically focused on predicting the actual RUL of the engine. Three different types of models were experimented with, and the results for each category of models are presented below.

1) Regression Models: In this category, various regression machine learning models, including linear regression, random forest, k-nearest neighbors, and others, were trained to predict the RUL. Different models exhibited varying performances, as shown in Table I, with the following observations.

The decision tree and the extra tree regressor overfitted, achieving 0 RMSE on the training set but recording higher RMSEs on the test set. Random forest had the best overall performance in this category, with an RMSE of 15.6 on the training set and 46.3 on the test set. However, the practicality of the random forest model was challenged by the high test set RMSE.

While the random forest exhibited better performance in the training set, the high test set RMSE raised practical concerns.

2) Deep Recurrent Models: In this category, various deep recurrent models, including Simple Recurrent Neural Networks (RNN), LSTM, GRU, and Bidirectional LSTM, were experimented with. These models were chosen due to their effectiveness with time series data, considering the time-dependent nature of sensor values. Sequences for training were generated using a window size of 50 cycles to ensure sufficient information for predicting the next values.

TABLE I
TRAIN AND TEST RMSE FOR REGRESSION MODELS

Model	Train RMSE	Test RMSE
Decision Tree	0.000000	69.070572
Extra Tree	0.000000	46.190967
Forest	15.626659	46.369789
XGB	28.174315	48.496991
KNR	40.501531	48.955520
SVM Reg	43.472257	48.873759
LReg	44.660360	48.399484
Ada Reg	47.671437	51.666636

Among these models, GRU and Bidirectional LSTM demonstrated superior performance. Bidirectional LSTM showed superiority in training RMSE, while GRU outperformed in test RMSE, as summarized in the figure below:

TABLE II
PERFORMANCE COMPARISON OF DEEP RECURRENT MODELS

Model	Train RMSE	Test RMSE
Bidirectional	12.292627	15.144974
GRU	12.880411	14.782531
RNN	13.034745	14.831060
LSTM	13.884804	15.078082

Overall, recurrent models proved effective in uncovering underlying patterns for predicting the RUL of an engine.

3) Deep Convolutional Models: In this category, various 1D and 2D CNN underwent training, and their performances were systematically compared. Among these models, CNN-4, considered as the best CNN model, exhibited exceptional results. It consisted of two convolutional layers and two max pooling layers. Specifically, the first convolutional layer featured 32 units, followed by a maximum pooling layer, and then another 1D convolutional layer with 64 units, succeeded by another maximum pooling layer. The output was subsequently flattened and directed to a fully connected layer housing 128 units, activated by the ReLU activation function. This model achieved a training RMSE of 11.09 and a test RMSE of 14.02, accompanied by an R-square value of 0.88.

Additionally, other CNN models were explored, including CNN-2, which comprised 1 convolutional layer and 1 max pool layer. Furthermore, the CNN-1+LSTM model was composed of 1 convolutional layer followed by 2 LSTM layers. The corresponding performances of CNN-2 and CNN-1+LSTM are detailed in Table III.

TABLE III
PERFORMANCE COMPARISON OF SELECTED CNN MODELS

Model	Train RMSE	Test RMSE
CNN-4	11.909235	14.023282
CNN-2	11.957053	16.370277
CNN-1+LSTM	12.477310	14.625450

The initial observations indicate that the proposed CNN model outperforms most of the reported values from other research studies, as indicated in Table IV.

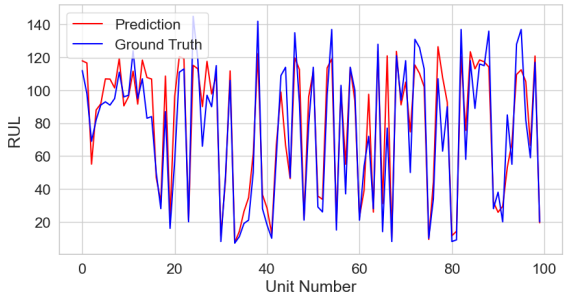

Fig. 4. Final Predictions on the Test Set

TABLE IV
COMPARISON OF VARIOUS CNN MODELS IN LITERATURE

Model	Test RMSE	R2	MAE
Proposed	**14.02**	**0.88**	**8.80**
Zheng et al. (2017)	16.42	NaN	NaN
Li et al. (2020)	NaN	0.78	13.45

The proposed CNN model showcased remarkable robustness in final predictions, as illustrated in Figure 4. The model's predictions exhibited a high degree of accuracy in closely tracking the actual values on the test set, thereby affirming its efficacy in predicting the RUL of the engine.

V. CONCLUSIONS AND FUTURE WORK

In this paper, the primary objective was to predict the Remaining Useful Life (RUL) of an aircraft engine, addressing it as a multi-faceted problem through binary classification and regression tasks. The project leveraged various machine learning techniques, including traditional regression models, deep recurrent and convolutional models, to explore the most effective approach for predicting engine failures and optimizing maintenance schedules.

The key contributions of this work are two-fold:

For the binary classification problem of predicting if an engine likely fails within the next month, our results show that the LSTM model of highest performance, achieving a precision of 0.96, recall of 0.88, and an F1 score of 0.92.

For the regression problem, different regression models and deep learning models were explored. Overall, the deep learning models, such as LSTM, Bidirectional LSTM, and GRU, and CNN models, exhibit high efficiency in predicting time-series data, showcasing their adaptability to the temporal nature of sensor values.

In particular, the proposed CNN model emerging as the best performer, achieving a training RMSE of 11.09 and a test RMSE of 14.02, demonstrating best efficiency in predicting the actual RUL of the engine. This superior performance suggests its potential for real-world applications in aviation predictive maintenance.

For future research, we propose to focus on further fine-tuning and optimizing the proposed CNN model to enhance its generalization capabilities and reduce test set RMSE. We also plan to exploit the more recent Transformer models [18]. Furthermore, we plan to explore ensemble models that combine the strengths of multiple approaches, such as other deep learning models, could potentially enhance predictive accuracy. Finally, we intend to investigate dynamic window sizes for recurrent models to adapt to varying lengths of historical data, ensuring effective utilization of information for predictions.

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